Mohammed Abdullah

Robotics Engineer (ROS / Embedded Systems)

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"Curiosity builds the path long before hard work walks it."

Summary

ECE student at the University College of Engineering, Osmania University, with interests in robotics, embedded systems, and sensor fusion. Focused on hardware–software integration and aiming for a career in robotics research. Actively contributes to open-source projects and participates in technology and science volunteering.

Skills

Programming Language
Python, C, C++
Robotics Framework & Middleware
ROS (Robot Operating System), Linux (Ubuntu)
Simulation & Physics Engines
Gazebo, PyBullet, MuJoCo, Modeling & Simulation
Embedded Systems & Hardware
Microcontrollers (ESP32, Arduino, STM32), Real-Time Operating Systems (FreeRTOS), Raspberry Pi, NVIDIA Jetson
Autonomy & Perception
Path Planning, Localization, OpenCV (Computer Vision), YOLO (Object Detection)
UAV & Drone Development
MissionPlanner, UAV Navigation, Drone Hardware Systems
Hardware Design & Tools
3D CAD (Autodesk Fusion 360), PCB Design (EasyEDA), Signal Processing, Electronics Repair, IoT
Soft Skills & Languages
Team Management, Public Speaking, Volunteering; English, Hindi, Urdu, Telugu, Arabic

Experience

Robotics Research Center — IIIT Hyderabad **Jan 2026 – Present**
Robotics Research Intern
Working under Prof. Hari Kumar on robotics systems research. Contributing to experiments, algorithm development, and system-level integration in robotics projects. Gaining hands-on experience in ROS, embedded systems, and applied automation.
robotics.iiit.ac.in

Keystone Robotix **Jul 2025 – Jan 2026**

Embedded Systems Intern

Worked on embedded programming, PCB development, and hardware–software integration for ongoing robotics projects. Contributed to robot programming, sensor interfacing, and testing on microcontrollers and processors. Skills involved include Python, C/C++, Arduino, OpenCV, Raspberry Pi, and Jetson Orin Nano. Exposure to ROS, YOLO, and basic 3D design.

keystonerobotix.com

Soham Academy (NGO). **Nov 2024 – Jun 2025**

Research Intern

Supported open-source robotics and electronics projects for community learning. Assisted the team in teaching robotics to government school students. Continued part-time volunteering for technical activities and STEM outreach.

sohamacademy.org

XplOrion **Feb 2025 – Present**

Founder

Founded a small initiative providing free/low-cost embedded systems solutions. Delivered four end-to-end projects for a client (Infinos Tech), including prototyping, electronics design, and final product development. Managing ongoing development as a side venture.

linkedin.com/company/xplorion

Projects

Smart Room Automation — ECE Dept., Osmania University **Aug 2025 – Nov 2025**

Developed a smart room system using ESP boards, MQTT, and touch-based interfaces. Designed custom enclosures in Fusion 360 and integrated IoT control for lighting and appliance automation.

Customized Voice Assistant (LLM-Free) **Sep 2025 – Oct 2025**

Built a voice assistant on Raspberry Pi 4 without using LLMs. Implemented wake-word detection, speech recognition, and lightweight NLP. Enabled smart home control with MQTT and fallback to web-based query responses. Demonstrated offline, local processing for home automation.

Drone Without Flight Controller **May 2025 – Jun 2025**

Designed and tested a drone controlled directly through Raspberry Pi 4, eliminating the need for a traditional flight controller. Worked on motor control, stabilization logic, and ROS-based system communication.

Wi-Fi & Bluetooth Jammer **Feb 2025 – Mar 2025**

Created a frequency-hopping jammer using ESP32 and NRF24 modules to generate noise signals across wireless bands. Implemented configurable modes for testing communication robustness.

Voice-Controlled Robotic Arm (4-DOF). **Nov 2024 – Dec 2024**

Built a 4-DOF robotic arm controlled through voice commands and Bluetooth. Implemented ROS-based communication, actuator control, and basic kinematic operation for simple task execution.

Education

University College of Engineering, Osmania University, Hyderabad **2024 – 2028 (Expected May)**
B.E. Electronics and Communication Engineering
CGPA: 7.5 (First two semesters combined)
Class representative for Semester 1; active in technical clubs, research activities, and department-level academic initiatives.
www.uceou.edu

Narayana Junior College, Nallakunta, Hyderabad **2022 – 2024**
Intermediate (MPC)
Telangana State Board (TGBIE)
CGPA: 9.5
Focused on core STEM subjects including Mathematics, Physics, Chemistry, and Arabic.
narayanajuniorcolleges.com

St. Andrews High School, Hyderabad **2009 – 2022**
Schooling (K–10)
Central Board of Secondary Education (CBSE)
CGPA: 9.1
Participated in public speaking, skating (school team), street-play events, and other extracurricular activities.
www.standrewssuchitra.com

Certifications & Achievements

SAEINDIA Southern Section, Autonomous Drone Development Challenge (ADDC 2026) **2026**
Best Design Report award, Overall Ranked 23rd among all teams.

Hobbies & Interests

Trekking & Outdoor Exploration (10+ treks across South India)
Weight Training